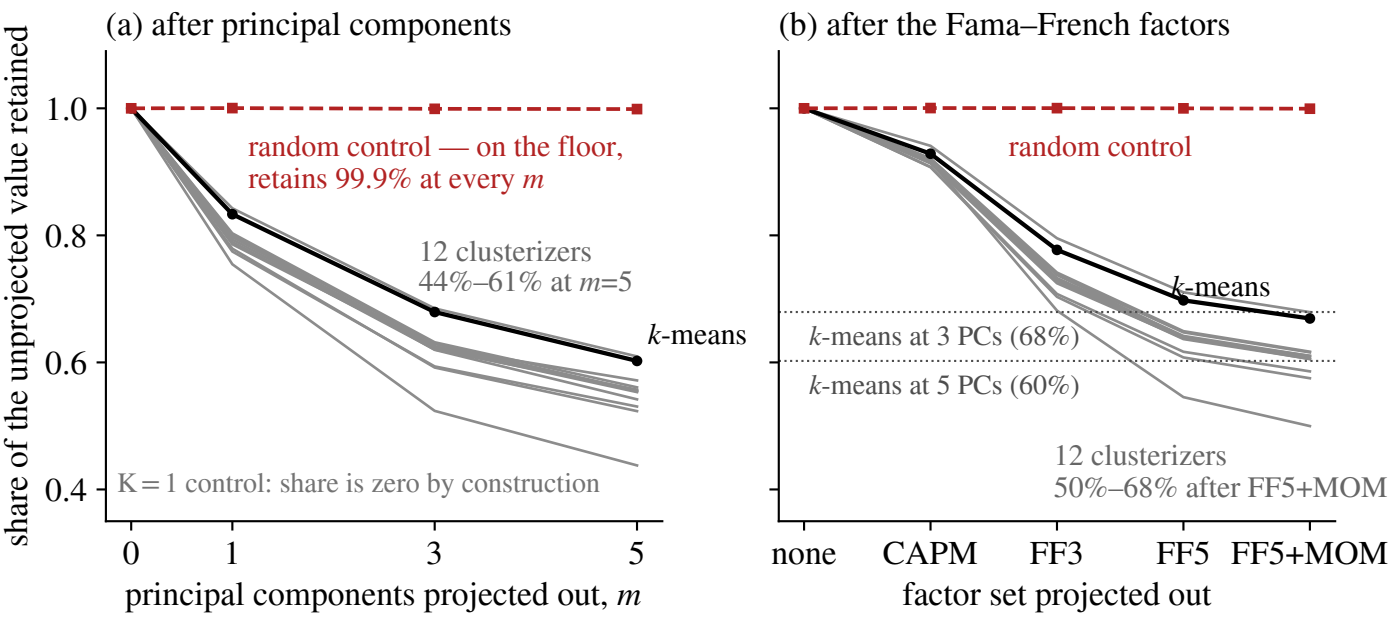
(a) after principal components
(b) after the Fama–French factors
share of the unprojected value retained
1.0
0.8
0.6
0.4
random control — on the floor,
retains 99.9% at every m
12 clusterizers
44%–61% at m=5
k-means
K = 1 control: share is zero by construction
0
1
3
5
principal components projected out, m
random control
k-means
k-means at 3 PCs (68%)
k-means at 5 PCs (60%)
12 clusterizers
50%–68% after FF5+MOM
none
CAPM
FF3
FF5
FF5+MOM
factor set projected out